

Accelerating Convergence

Theorem 2.8 indicates that it is rare to have the luxury of quadratic convergence. We now consider a technique called **Aitken's Δ^2 method** that can be **used to accelerate the convergence of a sequence that is linearly convergent**, regardless of its origin or application.

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|} = \lambda$$

Example with Fixed Point Iteration

- $f(x) = x^2 - x - 1$
- $x_{n+1} = 1 + \frac{1}{x_n}$
- $\epsilon = 10^{-7}$
- Start at $x_1 = 2$
- $r = \frac{1+\sqrt{5}}{2}$

$$\begin{aligned}
 x^2 - x - 1 &= 0 \\
 x^2 &= x + 1 \\
 x &= 1 + \frac{1}{x} \\
 x_{n+1} &= 1 + \frac{1}{x_n} \\
 \text{Pick } x_1 &= 2 \\
 x_2 &= 1 + \frac{1}{2} = 1.5
 \end{aligned}
 \qquad
 \begin{aligned}
 x^2 - x &= 1 \\
 x(x-1) &= 1 \\
 x &= \frac{1}{x-1} \\
 x_{n+1} &= \frac{1}{x_n - 1}
 \end{aligned}$$

n	x_n	Error (e_n)	Error Ratio
1	2	0.3819660113	-
2	1.5	-0.1180339887	-0.30901699437
3	1.66666666667	0.0486326779	-0.41202265919
4	1.6	-0.0180339887	-0.370820393225
5	1.625	0.0069660113	-0.386271242968682
6	1.61538461538	-0.0026493734	-0.380328609124042
7	1.61904761905	0.0010136303	-0.382592469458301
8	1.61764705882	-0.0003869299	-0.381726877989662
9	1.61818181818	0.0001478294	-0.382057366679621
10	1.61797752809	-0.0000564607	-0.381931120581252
11	1.61805555556	0.0000215668	-0.381979419745510
12	1.61802575107	-0.0000082377	-0.381960778685859
13	1.61803713528	0.0000031465	-0.381967798913113
14	1.61803278689	-0.0000012019	-0.381963571268055
15	1.61803444782	0.0000004591	-0.381966406379084
16	1.6180338134	-0.0000001753	-0.381967575106608

→ $\approx$

Aitken's Δ^2 Method

Suppose $\{p_n\}_{n=0}^{\infty}$ is a linearly convergent sequence with limit p . To motivate the construction of a sequence $\{\hat{p}_n\}_{n=0}^{\infty}$ that converges more rapidly to p than does $\{p_n\}_{n=0}^{\infty}$, let us first assume that the signs of $p_n - p$, $p_{n+1} - p$, and $p_{n+2} - p$ agree and that n is sufficiently large that

$$\frac{p_{n+1} - p}{p_n - p} \approx \frac{p_{n+2} - p}{p_{n+1} - p}.$$

Alexander Aitken (1895-1967) used this technique in 1926 to accelerate the rate of convergence of a series in a paper on algebraic equations [Ai]. This process is similar to one used much earlier by the Japanese mathematician Takakazu Seki Kowa (1642-1708).

Aitken's Δ^2 method is based on the assumption that the sequence $\{\hat{p}_n\}_{n=0}^{\infty}$, defined by

$$\hat{p}_n = p_n - \frac{(p_{n+1} - p_n)^2}{p_{n+2} - 2p_{n+1} + p_n}, \quad (2.14)$$

- Aitken, Alexander Craig. "XXV.—On Bernoulli's Numerical Solution of Algebraic Equations." *Proceedings of the Royal Society of Edinburgh* 46 (1927): 289-305.
- **Convergence Accelerator**
- Given a sequence of numbers with linear convergence use three consecutive numbers to predict new one using divided difference

n	x_n	Δx_n	$\Delta^2 x_n$
1	2	-0.5	0.6666666667
2	1.5		
3	1.6666666667	0.1666666666	
4	1.6		
5	1.625		

$$\hat{p}_n = x_n - \frac{(\Delta x_n)^2}{\Delta^2 x_n}$$

$$\hat{p}_n = x_n - \frac{(x_{n+1} - x_n)^2}{x_{n+2} - 2x_{n+1} + x_n}$$

$$\hat{p}_1 = (2) - \frac{(-0.5)^2}{(0.6666666667)}$$

$$\hat{p}_1 = 1.625$$

Example 1 The sequence $\{p_n\}_{n=1}^{\infty}$, where $p_n = \cos(1/n)$, converges linearly to $p = 1$. Determine the first five terms of the sequence given by Aitken's Δ^2 method.

Table 2.10

n	p_n	$\hat{p}_n$
1	0.54030	0.96178
2	0.87758	0.98213
3	0.94496	0.98979
4	0.96891	0.99242
5	0.98007	0.99451
6	0.98614	
7	0.98981	

$\hat{p}_n = p_n - \frac{(p_{n+1} - p_n)^2}{(p_{n+2} - 2p_{n+1} + p_n)}$

Forward Diff
Double Difference
 $\cos(1)$
 $\cos(\cos(1))$
 $\cos(\cos(\cos(1)))$
 $\downarrow g(p_n)$

Definition 2.13 For a given sequence $\{p_n\}_{n=0}^{\infty}$, the **forward difference** Δp_n (read “delta p_n ”) is defined by

$$\Delta p_n = p_{n+1} - p_n, \quad \text{for } n \geq 0.$$

Higher powers of the operator Δ are defined recursively by

$$\Delta^k p_n = \Delta(\Delta^{k-1} p_n), \quad \text{for } k \geq 2.$$

The definition implies that

$$\Delta^2 p_n = \Delta(p_{n+1} - p_n) = \Delta p_{n+1} - \Delta p_n = (p_{n+2} - p_{n+1}) - (p_{n+1} - p_n).$$

So $\Delta^2 p_n = p_{n+2} - 2p_{n+1} + p_n$, and the formula for $\hat{p}_n$ given in Eq. (2.14) can be written as

$$\hat{p}_n = p_n - \frac{(\Delta p_n)^2}{\Delta^2 p_n}, \quad \text{for } n \geq 0. \quad (2.15)$$

Theorem 2.14 Suppose that $\{p_n\}_{n=0}^{\infty}$ is a sequence that converges linearly to the limit p and that

$$\lim_{n \rightarrow \infty} \frac{p_{n+1} - p}{p_n - p} < 1.$$

Then the Aitken's Δ^2 sequence $\{\hat{p}_n\}_{n=0}^{\infty}$ converges to p faster than $\{p_n\}_{n=0}^{\infty}$ in the sense that

$$\lim_{n \rightarrow \infty} \frac{\hat{p}_n - p}{p_n - p} = 0.$$

Steffensen's Method

Johan Frederik Steffensen
(1873–1961) wrote an influential book entitled *Interpolation in* 1927.

By applying a modification of Aitken's Δ^2 method to a linearly convergent sequence obtained from fixed-point iteration, we can accelerate the convergence to quadratic. This procedure is known as Steffensen's method and differs slightly from applying Aitken's Δ^2 method directly to the linearly convergent fixed-point iteration sequence. Aitken's Δ^2 method constructs the terms in order:

$$\begin{aligned} p_0, \quad p_1 = g(p_0), \quad p_2 = g(p_1), \quad \hat{p}_0 = \{\Delta^2\}(p_0), \\ p_3 = g(p_2), \quad \hat{p}_1 = \{\Delta^2\}(p_1), \dots, \end{aligned}$$

where $\{\Delta^2\}$ indicates that Eq. (2.15) is used. Steffensen's method constructs the same first four terms, p_0, p_1, p_2 , and $\hat{p}_0$. However, at this step we assume that $\hat{p}_0$ is a better approximation to p than is p_2 and apply fixed-point iteration to $\hat{p}_0$ instead of p_2 . Using this notation, the sequence is

$$p_0^{(0)}, \quad p_1^{(0)} = g(p_0^{(0)}), \quad p_2^{(0)} = g(p_1^{(0)}), \quad p_0^{(1)} = \{\Delta^2\}(p_0^{(0)}), \quad p_1^{(1)} = g(p_0^{(1)}), \dots.$$

Every third term of the Steffensen sequence is generated by Eq. (2.15); the others use fixed-point iteration on the previous term. The process is described in Algorithm 2.6.

$$p^{(n+1)} = p^{(n)} - \frac{[g(p^{(n)}) - p^{(n)}]^2}{g(g(p^{(n)})) - 2g(p^{(n)}) + p^{(n)}}$$

- Steffensen, J. F. "Remarks on iteration." *Scandinavian Actuarial Journal* 1933.1 (1933): 64-72.
- "... our principle object being to show how the process of iteration which is often futile in its primitive form may be improved by a suitable combination of three consecutive values."
- "We shall therefore use as a working formula the approximation $x = x_v - \frac{(\Delta x_v)^2}{\Delta^2 x_v}$ where ... x_v may be any element of the sequence."

Steffensen's

To find a solution to $p = g(p)$ given an initial approximation p_0 :

INPUT initial approximation p_0 ; tolerance TOL ; maximum number of iterations N_0 .

OUTPUT approximate solution p or message of failure.

Step 1 Set $i = 1$.

Step 2 While $i \leq N_0$ do Steps 3–6.

Step 3 Set $p_1 = g(p_0)$; (Compute $p_1^{(i-1)}$.)
 $p_2 = g(p_1)$; (Compute $p_2^{(i-1)}$.)
 $p = p_0 - (p_1 - p_0)^2 / (p_2 - 2p_1 + p_0)$. (Compute $p_0^{(i)}$.)

Step 4 If $|p - p_0| < TOL$ then
 OUTPUT (p); (Procedure completed successfully.)
 STOP.

Step 5 Set $i = i + 1$.

Step 6 Set $p_0 = p$. (Update p_0 .)

Step 7 OUTPUT ('Method failed after N_0 iterations, $N_0 =$ ', N_0);
 (Procedure completed unsuccessfully.)
 STOP.

Illustration To solve $x^3 + 4x^2 - 10 = 0$ using Steffensen's method, let $x^3 + 4x^2 = 10$, divide by $x + 4$, and solve for x . This procedure produces the fixed-point method

$$g(x) = \left(\frac{10}{x + 4} \right)^{1/2}.$$

We considered this fixed-point method in Table 2.2 column (d) of Section 2.2.

Applying Steffensen's procedure with $p_0 = 1.5$ gives the values in Table 2.11. The iterate $p_0^{(2)} = 1.365230013$ is accurate to the ninth decimal place. In this example, Steffensen's method gave about the same accuracy as Newton's method applied to this polynomial. These results can be seen in the Illustration at the end of Section 2.4. $\square$

Table 2.11

$\longrightarrow g(x)$

k	$p_0^{(k)}$	$p_1^{(k)}$	$p_2^{(k)}$
0	1.5	1.348399725	1.367376372
1	1.365265224	1.365225534	1.365230583
2	1.365230013		

Example

$$f(x) = x^2 - x - 1$$

$$g(x) = 1 + \frac{1}{x}$$

$$\epsilon = 10^{-7}$$

Starting value 2

a	b	c	$\hat{p}$
2	1.5	1.666666667	1.625
1.625	1.61538461538	1.61904761905	1.61803713528
1.61803713528	1.61803278689	1.61803444782	1.61803398875
x_{13}	x_{14}	x_{15}	

Steffensen's Method 2.0

"To circumvent on the derivative calculation in Newton's [Method], Steffensen first coined the following second order convergent scheme: $x_{n+1} = x_n - \frac{(f(x_n))^2}{f(x_n + f(x_n)) - f(x_n)}$ by replacing the derivative in Newton's [Method] with the forward finite difference."

Kumar, Manoj, Akhilesh Kumar Singh, and Akanksha Srivastava. "Various Newton-type iterative methods for solving nonlinear equations." *Journal of the Egyptian Mathematical Society* 21.3 (2013): 334-339.

Out of Order

• Newton's Method

- $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$
- $\alpha = 2$

• Steffensen's Method

- $\hat{p}_n = x_n - \frac{(x_{n+1} - x_n)^2}{x_n - 2x_{n+1} + x_{n+2}}$
- $x_{n+1} = x_n - \frac{f(x_n)}{\frac{f(x_n + f(x_n)) - f(x_n)}{f(x_n)} - 1}$
- $\alpha = 2$

$$\alpha = \frac{\ln\left(\frac{e_{n+1}}{e_n}\right)}{\ln\left(\frac{e_n}{e_{n-1}}\right)}$$

n	$\hat{p}_n$	Error (e_n)
1	2	0.38196601125
2	1.625	0.00696601125
3	1.61803713528	0.00000314653
4	1.61803398875	0.00000000000001

$$\frac{\ln\left(\frac{e_3}{e_2}\right)}{\ln\left(\frac{e_2}{e_1}\right)} = \frac{\ln\left(\frac{0.00000314653}{0.00696601125}\right)}{\ln\left(\frac{0.00696601125}{0.38196601125}\right)} \approx 1.92$$

$$\frac{\ln\left(\frac{e_4}{e_3}\right)}{\ln\left(\frac{e_3}{e_2}\right)} = \frac{\ln\left(\frac{0.00000000000001}{0.00000314653}\right)}{\ln\left(\frac{0.00000314653}{0.00696601125}\right)} \approx 2.24$$

1. The following sequences are linearly convergent. Generate the first five terms of the sequence $\{\hat{p}_n\}$ using Aitken's Δ^2 method.
 - a. $p_0 = 0.5, \quad p_n = (2 - e^{p_{n-1}} + p_{n-1}^2)/3, \quad n \geq 1$
 - b. $p_0 = 0.75, \quad p_n = (e^{p_{n-1}}/3)^{1/2}, \quad n \geq 1$
3. Let $g(x) = \cos(x - 1)$ and $p_0^{(0)} = 2$. Use Steffensen's method to find $p_0^{(1)}$.
4. Let $g(x) = 1 + (\sin x)^2$ and $p_0^{(0)} = 1$. Use Steffensen's method to find $p_0^{(1)}$ and $p_0^{(2)}$.

Exercise Set 2.5, page 90

1. The results are listed in the following table.

	(a)	(b)	(c)	(d)
$\hat{p}_0$	0.258684	0.907859	0.548101	0.731385
$\hat{p}_1$	0.257613	0.909568	0.547915	0.736087
$\hat{p}_2$	0.257536	0.909917	0.547847	0.737653
$\hat{p}_3$	0.257531	0.909989	0.547823	0.738469
$\hat{p}_4$	0.257530	0.910004	0.547814	0.738798
$\hat{p}_5$	0.257530	0.910007	0.547810	0.738958

- *2. Newton's Method gives $p_6 = -0.1828876$ and $\hat{p}_6 = -0.183387$.
- *3. Steffensen's method gives $p_0^{(1)} = 0.826427$.
4. Steffensen's method gives $p_0^{(1)} = 2.152905$ and $p_0^{(2)} = 1.873464$.

Zeros of Polynomials and Müller's Method

A *polynomial of degree n* has the form

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where the a_i 's, called the *coefficients* of P , are constants and $a_n \neq 0$. The zero function, $P(x) = 0$ for all values of x , is considered a polynomial but is assigned no degree.

Algebraic Polynomials

Theorem 2.16 (Fundamental Theorem of Algebra)

If $P(x)$ is a polynomial of degree $n \geq 1$ with real or complex coefficients, then $P(x) = 0$ has at least one (possibly complex) root. ■

Carl Friedrich Gauss (1777–1855), one of the greatest mathematicians of all time, proved the Fundamental Theorem of Algebra in his doctoral dissertation and published it in 1799. He published different proofs of this result throughout his lifetime, in 1815, 1816, and as late as 1848. The result had been stated, without proof, by Albert Girard (1595–1632), and partial proofs had been given by Jean d’Alembert (1717–1783), Euler, and Lagrange.

Example 1 Determine all the zeros of the polynomial $P(x) = x^3 - 5x^2 + 17x - 13$.

Solution It is easily verified that $P(1) = 1 - 5 + 17 - 13 = 0$, so $x = 1$ is a zero of P and $(x - 1)$ is a factor of the polynomial. Dividing $P(x)$ by $x - 1$ gives

$$P(x) = (x - 1)(x^2 - 4x + 13).$$

To determine the zeros of $x^2 - 4x + 13$ we use the quadratic formula in its standard form, which gives the complex zeros

$$\frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(13)}}{2(1)} = \frac{4 \pm \sqrt{-36}}{2} = 2 \pm 3i.$$

Hence the third-degree polynomial $P(x)$ has three zeros, $x_1 = 1$, $x_2 = 2 - 3i$, and $x_3 = 2 + 3i$. ■

Horner’s Method

William Horner (1786–1837) was a child prodigy who became headmaster of a school in Bristol at age 18. Horner’s method for solving algebraic equations was published in 1819 in the *Philosophical Transactions of the Royal Society*.

To use Newton's method to locate approximate zeros of a polynomial $P(x)$, we need to evaluate $P(x)$ and $P'(x)$ at specified values. Since $P(x)$ and $P'(x)$ are both polynomials, computational efficiency requires that the evaluation of these functions be done in the nested manner discussed in Section 1.2. Horner's method incorporates this nesting technique, and, as a consequence, requires only n multiplications and n additions to evaluate an arbitrary n th-degree polynomial.

(Horner's Method)

Let

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0.$$

Define $b_n = a_n$ and

$$b_k = a_k + b_{k+1} x_0, \quad \text{for } k = n-1, n-2, \dots, 1, 0.$$

Then $b_0 = P(x_0)$. Moreover, if

$$Q(x) = b_n x^{n-1} + b_{n-1} x^{n-2} + \cdots + b_2 x + b_1,$$

then

$$P(x) = (x - x_0)Q(x) + b_0.$$

Paolo Ruffini (1765–1822) had described a similar method which won him the gold medal from the Italian Mathematical Society for Science. Neither Ruffini nor Horner was the first to discover this method; it was known in China at least 500 years earlier.

Example 2 Use Horner's method to evaluate $P(x) = 2x^4 - 3x^2 + 3x - 4$ at $x_0 = -2$.

	Coefficient of x^4	Coefficient of x^3	Coefficient of x^2	Coefficient of x	Constant term
$x_0 = -2$	$a_4 = 2$	$a_3 = 0$	$a_2 = -3$	$a_1 = 3$	$a_0 = -4$
		$b_4 x_0 = -4$	$b_3 x_0 = 8$	$b_2 x_0 = -10$	$b_1 x_0 = 14$
	$b_4 = 2$	$b_3 = -4$	$b_2 = 5$	$b_1 = -7$	$b_0 = 10$

So,

$$P(x) = (x + 2)(2x^3 - 4x^2 + 5x - 7) + 10. \quad \blacksquare$$

$$P(x) = (x - x_0)Q(x) + b_0,$$

where

$$Q(x) = b_n x^{n-1} + b_{n-1} x^{n-2} + \cdots + b_2 x + b_1,$$

differentiating with respect to x gives

$$P'(x) = Q(x) + (x - x_0)Q'(x) \quad \text{and} \quad P'(x_0) = Q(x_0).$$

When the Newton-Raphson method is being used to find an approximate zero of a polynomial, $P(x)$ and $P'(x)$ can be evaluated in the same manner.

Example 3 Find an approximation to a zero of

$$P(x) = 2x^4 - 3x^2 + 3x - 4,$$

using Newton's method with $x_0 = -2$ and synthetic division to evaluate $P(x_n)$ and $P'(x_n)$ for each iterate x_n .

Solution With $x_0 = -2$ as an initial approximation, we obtained $P(-2)$ in Example 1 by

$$\begin{array}{r|rrrrr} x_0 = -2 & 2 & 0 & -3 & 3 & -4 \\ & & -4 & 8 & -10 & 14 \\ \hline & 2 & -4 & 5 & -7 & 10 & = P(-2). \end{array}$$

Repeating the procedure to find x_2 gives

$$\begin{array}{r|rrrrr} -1.796 & 2 & 0 & -3 & 3 & -4 \\ & & -3.592 & 6.451 & -6.197 & 5.742 \\ \hline & 2 & -3.592 & 3.451 & -3.197 & 1.742 & = P(x_1) \\ & & -3.592 & 12.902 & -29.368 & \\ \hline & 2 & -7.184 & 16.353 & -32.565 & = Q(x_1) & = P'(x_1). \end{array}$$

So $P(-1.796) = 1.742$, $P'(-1.796) = Q(-1.796) = -32.565$, and

$$x_2 = -1.796 - \frac{1.742}{-32.565} \approx -1.7425.$$

Using Theorem 2.19 and Eq. (2.16),

$$Q(x) = 2x^3 - 4x^2 + 5x - 7 \quad \text{and} \quad P'(-2) = Q(-2),$$

so $P'(-2)$ can be found by evaluating $Q(-2)$ in a similar manner:

$$\begin{array}{r|rrrr}
 x_0 = -2 & 2 & -4 & 5 & -7 \\
 & & -4 & 16 & -42 \\
 \hline
 & 2 & -8 & 21 & -49 & = Q(-2) = P'(-2)
 \end{array}$$

and

$$x_1 = x_0 - \frac{P(x_0)}{P'(x_0)} = x_0 - \frac{P(x_0)}{Q(x_0)} = -2 - \frac{10}{-49} \approx -1.796.$$

In a similar manner, $x_3 = -1.73897$, and an actual zero to five decimal places is -1.73896 .

Note that the polynomial $Q(x)$ depends on the approximation being used and changes from iterate to iterate. ■